Document orientation

This document complements the Excel file 'laundry_datasets.xlsx' and is intended to guide the reader through the LCI data for use-phase/ laundry actions: dry cleaning, hand washing, machine washing, tumble drying and ironing. For each action it describes the parameters and the sources they are adopted from or based on.

The document is organized into nine sections:

Section 1-6: contain LCI data for the activities: dry cleaning, hand washing, machine washing, tumble drying, ironing and detergents. For each activity the related parameters, calculations and Ecoinvent/Biosphere3 exchanges are presented. Following information is presented:

'dataset description'	=	general comments and context about the dataset
'products'	=	specific activities/datasets representing variations of the action, defined by parameters such as material, temperature, or detergent type
'parameters'	=	input values used in the dataset/activity
'exchange formulas'	=	equations used to calculate exchange amounts
'exhange datasets'	=	details of the flows/exchanges, including the activities/datasets used and their source databases

A key assumption is that tap water and all wastewater, including wastewater with detergent, have a density of 1.

Section 7-9: contain additional information such as teachnical data, demostrative examples of calculations and references.

Overview

This document is structured according to the following listed sections:

Section	Products
1) Dry cleaning	▪ washing, dry cleaning
2) Hand washing	▪ hand washing, liquid detergent ▪ hand washing, powder detergent
3) Machine washing	▪ washing, 30°, average, liquid detergent ▪ washing, 30°, average, powder detergent ▪ washing, 40°, average, liquid detergent ▪ washing, 40°, average, powder detergent ▪ washing, 60°, average, liquid detergent ▪ washing, 60°, average, powder detergent
4) Tumble drying	▪ tumble drying, cotton, closet dry ▪ tumble drying, cotton, iron dry ▪ tumble drying, synthetics, closet dry
5) Ironing	▪ ironing, average, 1 min
6) Detergents	▪ liquid detergent ▪ powder detergent

7) Technical machine data	
8) Example calculations	
9) References	

determining variable

	parameter	unit	value	assumption
0A	garment weight	kg	1	Value set so that formulas are expressed per kilogram of garment.

1) Dry cleaning

dataset description

Dry cleaning is modelled with the use of perchloroethylene (PERC), which remains the most common chemical solvent applied in dry cleaning (Laitala, Klepp & Henry 2018). PERC is associated with both environmental and human health concerns and to reduce harmful emissions, modern dry cleaning machines employ technologies for improved solvent purification and vapor recovery systems. The main technologies used are carbon adsorbers and refrigerated condensers, which modern machines often use together. Together, they can reduce vapour emissions by up to 99%, after which the saturated activated carbon is disposed of as hazardous waste (Troynikov et al. 2016). For the dataset, dry cleaning has been represented in a simplified manner with inputs of PERC and electricity. The only output considered is the amount of PERC that becomes hazardous waste, as values for the amount of activated carbon used per kg of PERC and the saturation point of the activated carbon were not found. The model assumes that all PERC used in the process becomes hazardous waste.

products

- washing, dry cleaning

parameters

	parameter	unit	value	comment
1A	PERC amount	kg PERC / kg garment	0.0307	Value is an average of the PERC usage amounts (2-5.2 kg PERC per 100 kg laundry) reported by Keoleian et al. (1997). Uncertainty: Triangular (0.02, 0.0307, 0.052)
1B	electricity use	kWh / kg garment	0.586	Value adopted from Laitala, Klepp and Henry (2018).
1C	hazardous waste (percentage of PERC becoming hazardous waste)	%	100	It is assumed that the PERC is only used once and then discarded as hazardous waste (Troynikov et al. 2016).

exchange formulas

exchange	unit	formula
PERC amount	kg	$garment\ weight \cdot PERC\ amount$
electricity use	kWh	$garment\ weight \cdot electricity\ use$
hazardous waste	kg	$garment\ weight \cdot PERC\ amount \cdot \frac{hazardous\ waste}{100}$

exchange datasets

exchange name	reference product	unit	geography	sector	activity type	database

Tetrachloroethylene production	Tetrachloroethylene	kg	RoW	Chemicals	Transforming activity	Ecoinvent 3.10.1_cutoff
Market group for electricity, medium voltage	Electricity medium voltage	kWh	Europe without Switzerland	Electricity	Market group	Ecoinvent 3.10.1_cutoff
Market for hazardous waste, for incineration	Hazardous waste, for incineration	kg	Europe without Switzerland	Waste treatment & recycling	Market activity	Ecoinvent 3.10.1_cutoff

2) Hand washing

dataset description

Considered inputs are electricity to heat water, tap water and detergent. Only considered output is wastewater, which is caluclated based on a remaining moisture parameter of 50% (a bit higher than values for machine wash). This parameter describe how much water the garment uptakes based on the inital dry weight. Data on water and electricity use are based on German consumer behavior (Laitala et al. 2020). The LCI value for electricity use during hand washing, 0.165 kWh per kg of textile, reflects the observed distribution of wash temperatures: 52% lukewarm, 12% cold, 32% warm, and 4% hot, with an average temperature of 31.6 °C (Laitala et al. 2020). To determine the appropriate amount of detergent, Kruschwitz et al. (2014)´s recommendations for medium-soiled laundry was used.

products

- hand washing, liquid detergent
- hand washing, powder detergent

parameters

	parameter	unit	value	comment
2A	electricity use	kWh / kg garment	0.165	Value for german consumers based on Laitala et al. (2020).
2B	water use	L / kg garment	7.1	Value for german consumers based on Laitala et al. (2020).
2C	remaining moisture	%	50	Assuming that 50% of moisture remains in the clothes after the wash (value chosen to be a bit higher than the values from washing machine) and under the assumption that value refers to the weight of the clothes and not the water intake.
2D	liquid detergent amount	kg / kg garment	0.0169	Recommended dosage of liquid detergent for medium-soiled garments in water hardness <7.3–14 °dH. Values are based on Kruschwitz et al. (2014) and refers to machine washing conditions, applied here for hand washing. Uncertainty: Triangular(0.0116, 0.0169, 0.0284)
2E	powder detergent amount	kg / kg garment	0.0196	Recommended dosage of powder detergent for medium-soiled garments with water hardness 7.3-14 °dH. Value based on Kruschwitz et al. (2014) and refers to machine washing conditions, applied here for hand washing. Uncertainty: Triangular(0.0091, 0.0196, 0.0262)

exchange formulas

exchange	unit	formula
electricity use	kWh	$garment\ weight \cdot electricity\ use$
water use	kg	$garment\ weight \cdot water\ use$
wastewater	m3	$garment\ weight \cdot water\ use\ per\ wash \cdot 10^{-3} - garment\ weight \cdot \frac{remaining\ moisture}{100} \cdot 10^{-6}$

liquid detergent amount	kg	$garment\ weight \cdot\ liquid\ detergent\ use\ per\ wash$
powder detergent amount	kg	$garment\ weight \cdot\ powder\ detergent\ use\ per\ wash$

exchange datasets

exchange name	reference product	unit	geography	sector	activity type	database
Market group for electricity, low voltage	Electricity, low voltage	kWh	Europe without Switzerland	Electricity	Market group	Ecoinvent 3.10.1_cutoff
Market for tap water	Tap water	kg	Europe without Switzerland	Water supply	Market activity	Ecoinvent 3.10.1_cutoff
Market for wastewater, average	Wastewater, average	m3	Europe without Switzerland	Waste treatment & recycling	Market activity	Ecoinvent 3.10.1_cutoff
Detergents - see section 6) Detergents						

3) Machine washing

dataset description

The machine washing process accounts for the active electricity use of the machine during the selected program, the passive electricity use of when the machine is not in use, water intake, detergent use, and wastewater generation. Additionally, impacts related to acquiring and disposing of the washing machine are also considered.

It is assumed that the modern washing machine can take up to 8 kg of laundry. This assumption is based on Toft Ørbeck (2024), which notes that while earlier studies, such as Kruschwitz et al. (2014), reported average household washing machine capacities between 3 and 7 kg, more recent technological developments have increased these capacities. However, to maintain consistency with established consumer behavior data, the consumer loading tendencies for different washing programs (indicated by the load factors) observed in Kruschwitz et al. (2014) are used. This includes a tendency to underload cotton washes and overload wool washes.

Technical data for machine energy efficiency and program specific information is adopted from the AEG Washing Machine User Manual (LV7KR865E). Details on technical data and load factor assumptions can be found in section *7) Technical machine data*. The datasets (listed below under **products**) are based on the following programs: the ‘Wool/Silk 30°C’ program for 30°C washes, the ‘Synthetics 40°C’ program for 40°C washes, and the ‘Cotton 60°C’ program for 60°C washes. Detergent amounts are based on data from Kruschwitz et al. (2014) for medium-soiled laundry.

products

- washing, average, 30°, liquid detergent
- washing, average, 30°, powder detergent
- washing, average, 40°, liquid detergent
- washing, average, 40°, powder detergent
- washing, average, 60°, liquid detergent
- washing, average, 60°, powder detergent

parameters

For the parameter values which are dependent on program selection for the fibre type and dryness level, the parameter values will be presented as [] first providing the value for washing at 30°, then washing at 40°, and lastly washing at 60°.

	parameter	unit	value [30°, 40°, 60°]	comment
3A	lifetime	y	10	Value is from the documentation of the dataset “Market for washing machine” {Glo} from Ecoinvent.
3B	annual use	programs / y	220	Value adopted from AEG Washing Machine User Manual where it is used to calculate average electricity use per year.
3C	standard duration	min / program	195	The average wash duration was calculated by multiplying the duration of each program (cotton, synthetic, delicate, wool/silk) from the AEG Washing Machine User Manual by the number of times that program was selected (Kruschwitz et al., 2014). For the program selections ‘easy care’ and ‘mix’ the synthetic program duration was used. Uncertainty: Triangular(75, 195, 220)
3D	passive electricity use	W	0.3	It is assumed that the washing machine is connected to the electricity at all times and the passive electricity use is adopted from the AEG Washing Machine User Manual.
3E	standard load	kg / program	5.92	Value is based on the average load factor of 0.74 from Kruschwitz et al. (2014). Calculated as: 0.74*8 kg = 5.92 kg Uncertainty: Triangular(3.36, 5.92, 8.48)
3F	water use specific	L / program	[60, 60, 80]	Value adopted from AEG Washing Machine User Manual.
3G	load specific	kg / program	[1.605, 2.97, 5.44]	Calculated by multiplying the program specific average load factor reported by Kruschwitz et al. (2014) with the recommended load in the AEG washing machine manual. See the section containing the technical machine data for details.
3H	remaining moisture	%	[30, 35, 44]	Values adopted from the AEG Washing Machine User Manual. They are program specific and used under the assumption that the value of remaining most percentage refers to the weight of the clothes and not the water intake.
3I	electricity use specific	kWh / program	[0.3, 0.8, 1.3]	Program specific values adopted from the AEG Washing Machine User Manual.
3J	duration specific	h / program	[75/60, 140/60, 220/60]	Program specific values adopted from the AEG Washing Machine User Manual.
3K	liquid detergent amount	kg / kg garment	0.0169	Recommended dosage of liquid detergent for medium-soiled load in water hardness <7.3–14 °dH. Value based on Kruschwitz et al. (2014).
3L	powder detergent amount	kg / kg garment	0.0196	Recommended dosage of powder detergent for medium-soiled load in water hardness 7.3–14 °dH. Value based on Kruschwitz et al. (2014).

exchange formulas

exchange	unit	formula
Washing machine impact	unit	$\frac{garment\ weight}{lifetime \cdot annual\ use \cdot standard\ load}$
Electricity use (active)	kWh	$\frac{garment\ weight \cdot el\ use\ specific \cdot duration\ specific}{load\ specific}$
Electricity use (passive)	kWh	$\frac{passive\ energy\ use \cdot (lifetime \cdot 365.25 \cdot 24 \cdot 60 - lifetime \cdot annual\ use \cdot standard\ duration) \cdot \frac{1}{60} \cdot 10^{-3}}{lifetime} \cdot \frac{1}{annual\ use} \cdot \frac{1}{standard\ load} \cdot garment\ weight$

Water use	kg	$$garment\ weight \cdot water\ use\ specific \cdot \frac{1}{load\ specific}$$
Wastewater	m3	$$\frac{garment\ weight \cdot water\ use\ specific}{load\ specific} \cdot 10^{-3} - garment\ weight \cdot \frac{remaining\ moisture}{100} \cdot 10^{-6}$$
Liquid detergent amount	kg	$$garment\ weight \cdot liquid\ detergent\ amount$$
Powder detergent amount	kg	$$garment\ weight \cdot powder\ detergent\ amount$$

exchange datasets

exchange name	reference product	unit	geography	sector	activity type	database
Market for washing machine	Washing machine (includes disposal)	unit	GLO	Infrastructure & machinery	Market activity	Ecoinvent 3.10.1_cutoff
Market group for electricity, low voltage	Electricity, low voltage	kWh	Europe without Switzerland	Electricity	Market group	Ecoinvent 3.10.1_cutoff
Market for tap water	Tap water	kg	Europe without Switzerland	Water supply	Market activity	Ecoinvent 3.10.1_cutoff
Market for wastewater, average	Wastewater, average	m3	Europe without Switzerland	Waste treatment & recycling	Market activity	Ecoinvent 3.10.1_cutoff
Detergent - see section 6) Detergents						

4) Tumble drying

dataset description

The tumble drying process is modeled to account for the environmental impacts associated with the production and disposal of the dryer, as well as passive and active electricity consumption depending on the selected drying program. Technical specifications for energy efficiency and program details are taken from the AEG Dryer User Manual (LD8KR845E), see section 7) Technical machine data. In the absence of specific data on consumer tumble drying behavior, washing machine loading patterns were applied. Note that the value 'load specific' is based on the program´s recommended load factor. However, as with washing machines, it can be assumed that most users tend to underload dryers (Kruschwitz et al., 2014).

products

- tumble drying, cotton, closet dry
- tumble drying, cotton, iron dry
- tumble drying, synthetics, closet dry

parameters

For the parameter values which are dependent on program selection for the fibre type and dryness level, the parameter values will be presented as [] first providing the value for cotton closet dry, then cotton iron dry, and lastly synthetics closet dry cotton fibre.

	parameter	unit	value [cotton closet dry, cotton iron dry,	comment

			synthetics closet dry]	
4A	lifetime	y	12.5	Value is from the documentation of the dataset "Market for dryer" {Glo} from Ecoinvent.
4B	annual use	programs / y	160	Value adopted from AEG Dryer User Manual where it is used to calculate average electricity use per year.
4C	standard duration	min / program	108.9	Value is from Laitala et al (2020) and represents the average tumble drying time {Germany}. Uncertainty: Triangular(55, 108.9, 154)
4D	passive electricity use	W	0.13	It is assumed that the dryer is connected to the electricity at all times and the passive electricity use is adopted from the AEG Dryer User Manual.
4E	standard load	kg / program	5.92	Assuming that most people tend to underload dryers (as is true for washing machines), the value is based on the average load factor of 0.74 from Kruschwitz et al. (2014). Calculated as: 0.74*8 kg = 5.92 kg Uncertainty: Triangular(2.20, 5.92, 6.05)
4F	electricity use specific	kWh / program	[1.99,1.51, 0.75]	Data from AEG Dryer User Manual for 1000 spins program for cotton and 800 spins for synthetics.
4G	duration specific	h / program	[154/60, 118/60, 67/60]	Data from AEG Dryer User Manual for 1000 spins program for cotton and 800 spins for synthetics.
4H	load specific	kg / program	[8, 8, 3.5]	Value is set to the program recommended one from AEG Dryer User Manual, since no data is available on dyer loading tendencies.
4I	remaining moisture after wash	%	[44, 44, 35]	Values adopted from the AEG Washing Machine User Manual. They are program specific and used under the assumption that the value of remaining most percentage refers to the weight of the clothes and not the water intake.

exchange formulas

exchange	unit	formula
Tumble dryer machine impact	unit	$\frac{1}{lifetime} \cdot \frac{1}{annual\ use} \cdot \frac{1}{standard\ load} \cdot garment\ weight \left(1 + \frac{remaining\ moisture}{100}\right)$
Electricity use (active, closet dry)	kWh	$garment\ weight \cdot (1 + \frac{remaining\ moisture}{100}) \cdot El\ use\ specific\ closet\ dry \cdot Duration\ specific\ closet\ dry \cdot \frac{1}{Load\ specific\ closet\ dry}$
Electricity use (active, iron dry)	kWh	$garment\ weight \cdot (1 + \frac{remaining\ moisture}{100}) \cdot El\ use\ specific\ Iron\ dry \cdot Duration\ specific\ Iron\ dry \cdot \frac{1}{Load\ specific\ Iron\ dry}$
Electricity use (passive)	kWh	$\frac{passive\ energy\ use \cdot (lifetime \cdot 365.25 \cdot 24 \cdot 60 - lifetime \cdot annual\ use \cdot standard\ duration) \cdot \frac{1}{60} \cdot 10^{-3}}{lifetime} \cdot \frac{1}{annual\ use} \cdot \frac{1}{standard\ load} \cdot garment\ weight$

exchange datasets

exchange name	reference product	unit	geography	sector	activity type	database
Market for dryer	Dryer (includes disposal)	unit	GLO	Infrastructure & machinery	Market activity	Ecoinvent 3.10.1_cutoff
Market group for electricity, low voltage	Electricity, low voltage	kWh	Europe without Switzerland	Electricity	Market group	Ecoinvent 3.10.1_cutoff

5) Ironing

dataset description

The ironing process is modeled per minute of residential use and includes the impacts of producing and disposing of an electric iron, as well as electricity consumption, estimated at 0.027 kWh per minute (Sandin et al. 2019). According to Eurostat HETUS data (estat_tus_20age.tsv, 2020), which collects time-use information from five European countries (BG, EE, AT, FI, RS), a value of 5.87 minutes per day spent ironing was used. Mass allocation was not used for this process. Since Ecoinvent lacks data for irons or steamers, the impact of an electric kettle was used as a proxy. Both appliances are small household devices designed to heat water, and ironing and steaming are assumed to be similar in their resource use.

Note: Average ironing time per garment can be modeled following Adams et al. (2025), with typical values such as 2.6 minutes for a t-shirt and 4.3 minutes for pants.

products

- ironing, average, 1 min

parameters

	parameter	unit	value	comment
5A	time spend	min	1	Value set so that formulas are expressed per minute of ironing.
5B	lifetime of electric kettle	y	4	Value is from the documentation of the dataset “market for electric kettle" {GLO} from Ecoinvent. Since no data for an iron or steamer were available, an electric kettle was used as a proxy, based on the assumption that both are small household appliances designed to heat water and thus have comparable environmental impacts.
5C	electricity use	kWh/min	0.027	Value is adopted for minute of ironing from Sandin et al. (2019), assuming an average iron power of 1600 kW.
5D	daily time spend on ironing	min	5,87	Value based on Eurostat (estat_tus_20age.tsv) HETUS data (2020) for the category ‘CARE FOR TEXTILES’ (331 Laundry, 332 Ironing, 339 Other textile care). Since ironing specific data are not reported separately, the combined category is used as a proxy. An average across the five European countries (BG, EE, AT, FI, RS) is applied. To better approximate actual ironing time, 33% of the reported value is used. Calculated as: 17.8 · 0.33 ≈ 5.87 minutes/day.

exchange formulas

exchange	unit	formula
Kettle machine impact	unit	$\frac{time\ spend}{lifetime\ of\ electric\ kettle\ \cdot\ daily\ time\ spend\ on\ ironing\ \cdot\ 7\ \cdot\ 52}$
Electricity use	kWh	$time\ spend \cdot\ electricity\ use$

exchange datasets

exchange name	reference product	unit	geography	sector	activity type	database
Market for electric kettle	Electric kettle (includes disposal)	unit	GLO	Infrastructure & machinery	Market activity	Ecoinvent 3.10.1_cutoff
Market group for electricity, low voltage	Electricity, low voltage	kWh	Europe without Switzerland	Electricity	Market group	Ecoinvent 3.10.1_cutoff

6) Detergents

No general composition of detergent types for residential laundry was found in Ecoinvent and activities for liquid and powder detergent have therefore been created.

liquid detergent - dataset description

The LCI dataset for liquid detergent is based on Toft Ørbeck (2024) and Sandin et al. (2019). It represents the resource inputs per kilogram of detergent and includes detergent ingredients, process energy, and packaging materials. The dataset does not account for the disposal of packaging.

products

- liquid detergent

Liquid detergent								
detergent ingredients	dataset in model	reference product	geography	sector	activity type	database	quantity	unit
	Input							
Alkyl sulphate	Market for alkyl sulphate (C12-14)	Alkyl sulphate (C1214)	GLO	Chemicals	Market activity	Ecoinvent 3.10.1_cutoff	0.1038	kg
Citric acid	Citric acid production	Citric acid	RER	Chemicals	Transforming activity	Ecoinvent 3.10.1_cutoff	0.0228	kg
Enzymes	Enzymes production	Enzymes	RER	Chemicals	Transforming activity	Ecoinvent 3.10.1_cutoff	0.0058	kg
Glycerine	Market for glycerine	Glycerine	RER	Chemicals	Market activity	Ecoinvent 3.10.1_cutoff	0.0285	kg
Non-ionic surfactant	Market for non-ionic surfactant	Non-ionic surfactant	GLO	Chemicals	Market activity	Ecoinvent 3.10.1_cutoff	0.0591	kg
Polyethylene	Market for polyethylene, linear low density, granulate	Polyethylene, linear low density, granulate	GLO	Chemicals	Market activity	Ecoinvent 3.10.1_cutoff	0.0466	kg
Soap	Soap production	Soap	RER	Chemicals	Transforming activity	Ecoinvent 3.10.1_cutoff	0.0241	kg
Sodium hydroxide	Market for sodium hydroxide, without water, in 50% solution state	Sodium hydroxide, without water, in 50% solution state	RER	Chemicals	Market activity	Ecoinvent 3.10.1_cutoff	0.0231	kg
Water	Market for water, deionised	water, deionised	Europe without Switzerland	Water supply	Market activity	Ecoinvent 3.10.1_cutoff	0.7022	kg
Electricity	Market group for electricity, medium voltage	Electricity, medium voltage	RER	Electricity	Market group	Ecoinvent 3.10.1_cutoff	0.25	kWh
Packaging materials								
HDPE bottle	Market for polyethylene, high density, granulate	Polyethylene, high density, granulate	GLO	Chemicals	Market activity	Ecoinvent 3.10.1_cutoff	0.0466	kg
PP cork	Market for polypropylene, granulate	Polypropylene, granulate	GLO	Chemicals	Market activity	Ecoinvent 3.10.1_cutoff	0.0101	kg
Label	Market for printed paper	Printed paper	GLO	Electronics, pulp & paper	Market activity	Ecoinvent 3.10.1_cutoff	0.00126	kg
	Output							
	Liquid detergent (density 0.95 kg/L)						1	kg

powder detergent - dataset description

The LCI dataset for powder detergent is based on Toft Ørbeck (2024), which builds on Moazzem et al. (2021). It has been updated to include packaging materials (Saouter & Van Hoof 2002; Roos et al. 2015) and to replace LAS-pc, a detergent ingredient largely phased out in Europe and assumed to be substituted with ethoxylated alcohols (Roos et al. 2015). The dataset covers resource inputs and production related emissions per kilogram of powder detergent. It does not include packaging disposal.

products

- powder detergent

Powder detergent								
detergent ingredients	dataset in model	reference product	geography	sector	activity type	database	quantity	unit
	Input							
AE11-PO	Market for ethoxylated alcohol (AE11)	Ethoxylated alcohol (AE11)	GLO	Chemicals	Market activity	Ecoinvent 3.10.1_cutoff	0.02	kg
AE7-pc	Market for ethoxylated alcohol (AE7)	Ethoxylated alcohol (AE7)	RER	Chemicals	Market activity	Ecoinvent 3.10.1_cutoff	0.04	kg
LAS-pc							0.078	kg
Citric acid	Market for citric acid	Citric acid	GLO	Chemicals	Transforming activity	Ecoinvent 3.10.1_cutoff	0.052	kg
Na-Silicate powder	Market for layered sodium silicate, SKS-6, powder	Layered sodium silicate, SKS-6, powder	GLO	Chemicals	Market activity	Ecoinvent 3.10.1_cutoff	0.03	kg
Zeolite	Market for zeolite, powder	Zeolite, powder	GLO	Chemicals	Market activity	Ecoinvent 3.10.1_cutoff	0.201	kg
Sodium carbonate	Market for soda ash, dense	Soda ash, dense	GLO	Chemicals	Market activity	Ecoinvent 3.10.1_cutoff	0.17	kg
Perborate monohydrate	Market for sodium perborate, monohydrate, powder	Sodium perborate, monohydrate, powder	GLO	Chemicals	Market activity	Ecoinvent 3.10.1_cutoff	0.087	kg
Perborate tetrahydrate	Market for sodium perborate, tetrahydrate, powder	Sodium perborate, tetrahydrate, powder	RER	Chemicals	Market activity	Ecoinvent 3.10.1_cutoff	0.115	kg
Antifoam S1.2-3522	Not available in ecoinvent						0.005	kg
FWA DAS-1	Not available in ecoinvent						0.002	kg
Polyacrylate	Not available in ecoinvent						0.04	kg
Protease	Not available in ecoinvent						0.014	kg
Sodium sulfate	Market for sodium sulfate, anhydrite	Sodium sulfate, anhydrite	RER	Chemicals	Market activity	Ecoinvent 3.10.1_cutoff	0.004	kg
Water	Market for water, completely softened	Water, completely softened	RER	Water supply	Market activity	Ecoinvent 3.10.1_cutoff	0.142	kg
Energy	Market group for electricity, medium voltage	Electricity, medium voltage	RER	Electricity	Market group	Ecoinvent 3.10.1_cutoff	0.69	kWh
Packaging materials								
Paper woody U B250 (1998)	Market for kraft paper	kraft paper	RER	Pulp & Paper	Market Activity	Ecoinvent 3.10.1_cutoff	0.0217	kg
Corrugated cardboard	Market for corrugated board box	corrugated board box	RER	Pulp & Paper	Market Activity	Ecoinvent 3.10.1_cutoff	0.1082	kg

HDPE B250 (barrier)	Polyethylene production, high density, granulate	polyethylene, high density, granulate	RER	Chemicals	Market Activity	Ecoinvent 3.10.1_cutoff	0.0081	kg
	Output							
	Powder detergent						1	kg
	Emissions to air CO2	Carbon dioxide, fossil				Biosphere3	0.133	kg
	Emissions to air CO	Carbon monoxide, fossil				Biosphere3	0.00006	kg
	Emissions to air SOx	Sulfur dioxide				Biosphere3	0.000696	kg
	Emissions to air NOx	Nitrogen oxides				Biosphere3	0.000329	kg
	Emissions to air CxHy	Not available in Biosphere3					0.000109	kg
	Emissions to air - particles/dust	Not available in Biosphere3					0.000176	kg
	Emissions to water BOD	BOD5, Biological Oxygen Demand				Biosphere3	0.000049	kg
	Emissions to water COD	COD, Chemical Oxygen Demand				Biosphere3	0.000101	kg

7) Technical machine data

Machine Washing

AEG Washing machine, User Manual, LD8KR845E					
	Load	Electricity consumption	Water consumption	Duration	Remaining moisture
Wool/Silk 30°C	1.5 kg	0.3 kWh	60 L	75 min	30%
Synthetics 40°C	3 kg	0.8 kWh	60 L	140 min	35%
Cottons 60°C	8 kg	1.3 kWh	80 L	220 min	44%
Passive electricity use of 0.3 W					

Assumed load per program			
	Average load factor (Kruschwitz et al., 2014)	Max weight (AEG Washing Machine User Manual)	Assumed load per program
Wool	1.07	1.5 kg	1.07*1.5= 1.605 kg
Synthetics	0.99	3 kg	0.99*3=2.97 kg
Cotton	0.68	8 kg	0.68*8=5.44 kg

Tumble Drying

AEG Dryer, User Manual, LD8KR845E				
	Load	Spun at	Drying time	Electricity consumption
Cotton, closet dry	8 kg	1000 rounds/min	154 min	1.99 kWh
Cotton, iron dry	8 kg	1000 rounds/min	118 min	1.51 kWh
Synthetics, closet dry	3.5 kg	800 rounds/min	67 min	0.75 kWh
Passive electricity use of 0.13 W				
All drying programs are eco				

8) Example calculations

This section provides detailed explanations of the formulas used in activities in section 1) - 6). It includes step-by-step calculations with units and written explanations to support understanding of how each formula is applied.

Determining variable

Parameters

$0A = garment\ weight\ [kg\ garment] = 1$

1) Dry cleaning

Parameters

$1A = PERC\ amount\ [kg\ PERC\ /\ kg\ garment] = 0.0307$
$1B = electricity\ use\ [kWh\ /\ kg\ garment] = 0.586$
$1C = hazardous\ waste\ (percentage\ of\ PERC\ becoming\ hazardous\ waste)\ [\%] = 100$

Activities

PERC use

This equation calculates the amount of the chemical solvent PERC that is used in the dry cleaning process per kg garments.

$$garment\ weight\ [kg\ garment] \cdot PERC\ amount\ \left[\frac{kg\ PERC}{kg\ garment}\right] = PERC\ amount\ [kg]$$

$$0A \cdot 1A = 1 \cdot 0.0307 = 0.0307$$

Electricity use

This equation calculates the amount of electricity consumed in the dry cleaning process per kg garments.

$$garment\ weight\ [kg\ garment] \cdot electricity\ use\ \left[\frac{kWh}{kg\ garment}\right] = electricity\ use\ [kWh]$$

$$1 \cdot 0.586 = 0.586$$

Hazardous waste generation

This equation calculates the amount of the chemical solvent PERC from the dry cleaning process that becomes hazardous waste per kg garments.

$$garment\ weight\ [kg\ garment] \cdot PERC\ amount\ \left[\frac{kg\ PERC}{kg\ garment}\right] \cdot \frac{hazardous\ waste\ [\%]}{100} = hazardous\ waste\ [kg]$$

$$1 \cdot 0.0307 \cdot \frac{100}{100} = 0.0307$$

2) Hand washing

Example for 'hand washing, liquid detergent'.

Parameters

$2A = P\ electricity\ use\ [kWh\ /\ kg\ garment] = 0.165$
$2B = water\ use\ [L\ /\ kg\ garment] = 7.1$
$2C = remaining\ moisture\ [\%] = 50$
$2D = liquid\ detergent\ use\ [kg\ /\ kg\ garment] = 0.0169$

Activities

Electricity use / kg garment

This equation calculates the electricity consumed per kg of garments during hand washing.

$$garment\ weight\ [kg\ garment] \cdot electricity\ use\ \left[\frac{kWh}{kg\ garment}\right] = electricity\ use\ [kWh]$$

$$1 \cdot 0.165 = 0.165$$

Water use / kg garment
This equation calculates the amount of water used per kg of garments during hand washing.

$$garment\ weight\ [kg\ garment] \cdot water\ use\ \left[\frac{L}{kg\ garment}\right] = \ water\ use\ [kg]$$

$$1\ \cdot 7.1 = \ 7.1$$

Wastewater / kg garment
This equation calculates the amount of wastewater generated per kg of garments. It accounts for water used minus the water retained in the garments after washing (remaining moisture).

$$garment\ weight[kg\ garment] \cdot water\ use\ per\ wash\ \left[\frac{L}{kg\ garment}\right] \cdot 10^{-3} - garment\ weight \cdot \frac{remaining\ moisture\ [\%]}{100} \cdot 10^{-6} = wastewater\ [m3]$$

$$1\ \cdot 7.1\ \ \cdot 10^{-3} - 1 \cdot \frac{50}{100} \cdot 10^{-6}\ \ = 0.0070995$$

Detergent use / kg garment
This equation determines the detergent usage. Replacing the liquid detergent amount (2D) with the powder detergent amount (2E) allows the equation to be applied to either detergent type.

$$garment\ weight\ [kg\ garment] \cdot\ liquid\ detergent\ use\ \left[\frac{kg}{kg\ garment}\right] = \ detergent\ use\ [kg]$$

$$1\ \cdot 0.0169\ = \ 0.0169$$

3) Machine washing

The calculated values are provided as an example for the activity 'washing, average, 30°, liquid detergent'.

Parameters

$3A\ = \ lifetime\ [y] = \ 10$
$3B\ = \ annual\ use\ [programs\ /\ y] = \ 220$
$3C\ = standard\ duration\ [min/\ program]\ = \ 195$
$3D\ = passive\ electricity\ use\ [W]\ = \ 0.3$
$3E\ = \ standard\ load\ [kg\ /\ program] = \ 5.92$
$3F\ = \ water\ use\ specific\ [L\ /\ program]\ = \ 60$
$3G\ = load\ specific\ [kg/\ program]\ = \ 1.605$
$3H\ = remaining\ moisture\ [\%]\ = \ 30$
$3I\ = \ electricity\ use\ specific\ [kWh\ /\ program] = \ 0.3$
$3J\ = \ duration\ specific\ [h\ /\ program]\ = \ \frac{75}{60}$
$3K\ = liquid\ detergent\ use\ [kg/\ kg\ garment]\ = \ 0.0169$

Activities

Washing machine impact / kg garment
This equation calculates the portion of the washing machine´s total environmental impact that is allocated to each kilogram of washed garments.

$$\frac{unit}{lifetime[y]} \cdot \frac{1}{annual\ use\ \left[\frac{programs}{y}\right]} \cdot \frac{1}{standard\ load\ \left[\frac{kg}{program}\right]} \cdot garment\ weight\ [kg\ garment] = [unit]$$

$$\frac{1}{10} \cdot \frac{1}{220} \cdot \frac{1}{5.92} \cdot 1 = \ 7.678132678132678 \cdot 10^{-5}$$

Electricity use (active) / kg garment
This equation calculates the active electricity use of the machine for the 30° washing program.

$$\frac{garment\ weight\ [kg\ garment] \cdot\ electricity\ use\ specific\ \left[\frac{kWh}{kg\ garment}\right] \cdot\ duration\ specific \left[\frac{h}{kg\ garment}\right]}{load\ specific\ \left[\frac{kg}{program}\right]}$$
$$=\ electricity\ use\ [kWh]$$
$$\frac{1\ \cdot 0.3\ \cdot \frac{75}{60}}{1.605} = 0.2336448598130841$$

Electricity use (passive) / kg garment

This equation calculates the electricity consumed by the washing machine during periods it is plugged in but not in active use.

Lifetime of machine

$$lifetime[y] \cdot 365.25 \left[\frac{d}{y}\right] \cdot 24 \left[\frac{h}{d}\right] \cdot 60 \left[\frac{min}{h}\right] =\ lifetime\ [min]$$
$$X1 =\ 10\ \cdot 365.25 \cdot 24 \cdot 60 =\ 5\ 259\ 600$$

Average active use during lifetime

$$lifetime[y] \cdot annual\ use \left[\frac{programs}{y}\right] \cdot standard\ duration \left[\frac{min}{program}\right]$$
$$=\ active\ use\ during\ lifetime\ [min]$$
$$X2 =\ 10\ \cdot 220 \cdot 195 =\ 429\ 000$$

Average passive lifetime of machine

$$X3 =\ X1\ -\ X2\ =\ \ 4\ 830\ 600$$

Passive electricity use during lifetime

$$passive\ electricity\ use\ [W] \cdot XC[min] \cdot 60^{-1} \left[\frac{h}{min}\right] \cdot 10^{-3} \left[\frac{kW}{W}\right]$$
$$=\ passive\ electricity\ use\ during\ lifetime\ [kWh]$$
$$X4 =\ 0.3\ \cdot\ X3 \cdot\ 60^{-1} \cdot 10^{-3} =\ 24.153$$

Passive electricity use per kg garment

$$\frac{X4\ [kWh]}{lifetime[y]} \cdot \frac{1}{annual\ use\ [\frac{programs}{y}]} \cdot \frac{1}{standard\ load\ [\frac{g}{program}]} \cdot garment\ weight\ [kg\ garments]$$
$$=\ passive\ electricity\ use\ [kWh]$$
$$\frac{X4}{10} \cdot \frac{1}{220} \cdot \frac{1}{5.92} \cdot 1 = 0.001854499385749386$$

Water use / kg garment

This equation calculates the water use of the machine for the 30° washing program.

$$\frac{garment\ weight\ [kg\ garment] \cdot\ water\ use\ specific\ \left[\frac{L}{kg\ garment}\right]}{load\ specific\ \left[\frac{kg}{program}\right]} \cdot 1[\frac{kg\ water}{L\ water}] = water\ use\ [kg]$$
$$\frac{1\ \cdot 60}{1.605} \cdot 1 = 37.38317757009346$$

Wastewater / kg garment

This equation calculates the wastewater generation for the 30° washing program.

$$\left(\frac{garment\ weight\ [kg\ garment] \cdot\ water\ use\ specific\ \left[\frac{L}{kg\ garment}\right]}{load\ specific\ \left[\frac{kg}{program}\right]}\right) \cdot 10^{-3} \left[\frac{m^3\ water}{L\ water}\right]$$
$$- \left(garment\ weight[kg\ garments] \cdot \frac{remaining\ moisture[\%]}{100} \cdot 10^{-3} \left[\frac{m^3}{kg}\right]\right)$$
$$= wastewater\ [m^3]$$

$$\left(\frac{1\ \cdot 60}{1.605}\right) \cdot 10^{-3} - \left(1 \cdot \frac{30}{100} \cdot 10^{-3}\right) = 0.03708317757009346$$

Detergent use / kg garment

This equation determines the detergent usage. Replacing the liquid detergent amount (3K) with the powder detergent amount (3L) allows the equation to be applied to either detergent type.

$$garment\ weight[kg\ garment] \cdot liquid\ detergent\ amount \left[\frac{kg}{kg\ garment}\right] = detergent\ use\ [kg]$$

$$1 \cdot 0.0169 = 0.0169$$

4) Tumble drying

The calculated values are provided as an example for the product 'tumble drying, cotton, closet dry'.

Parameters

$4A\ = lifetime\ [y] = 12.5$
$4B\ = annual\ use\ [programs\ /\ y] = 160$
$4C\ = standard\ duration\ [min/\ program]\ = 108.9$
$4D\ = passive\ electricity\ use\ [W]\ = 0.13$
$4E\ = standard\ load\ [kg\ /\ program] = 5.92$
$4F\ = electricity\ use\ specific\ [kWh\ /\ program]\ = 1.99$
$4G\ = duration\ specific\ [h\ /\ program]\ = \frac{154}{60}$
$4H\ = load\ specific\ [kg/\ program]\ = 8$

$4I\ = remaining\ moisture\ after\ wash\ [\%]\ = 44$

Activities

Tumble dryer machine impact / kg garment

This equation calculates the portion of the tumble dryer´s environmental impact allocated per kilogram of garments, accounting for remaining moisture after washing.

$$\frac{unit}{lifetime[y]} \cdot \frac{1}{annual\ use\ \left[\frac{programs}{y}\right]} \cdot \frac{1}{standard\ load\ \left[\frac{kg}{program}\right]} \cdot garment\ weight\ [kg\ garment]$$
$$\cdot \left(1 + \frac{remaining\ moisture\ after\ wash\ [\%]}{100}\right) = [unit]$$

$$\frac{1}{12.5} \cdot \frac{1}{160} \cdot \frac{1}{5.92} \cdot 1 \cdot \left(1 + \frac{44}{100}\right] = 0.000121621621621621$$

Electricity use (active - closet dry) / kg garment

This equation calculates the electricity use of the tumble dryer allocated per kilogram of garment, based on the wet weight.

$$\frac{garment\ weight\ [kg\ garment] \cdot \left(1 + \frac{remaining\ moisture\ after\ wash\ [\%]}{100}\right) \cdot electricity\ use\ specific\ \left[\frac{kWh}{program}\right] \cdot\ duration\ specific\ \left[\frac{h}{program}\right]}{load\ specific\ \left[\frac{kg}{program}\right]}$$
$$= \text{electricity use [kWh]}$$

$$1 \cdot (1 + \frac{44}{100}) \cdot 1.99 \cdot \frac{154}{60} \cdot \frac{1}{8} = 0.91938$$

Electricity use (passive) / kg garment
This calculation estimates the passive electricity use of the tumble dryer per kilogram of garment. Passive electricity is the energy consumed while the appliance is plugged in but not actively drying. See activity Electricity use (passive) / kg garment under 3) Machine washing for detailed explanation.

$$\text{garment weight } [kg\ garment] \cdot passive\ electricity\ use[W] \cdot \frac{\left(lifetime[y] \cdot 365.25 \left[\frac{d}{y}\right] \cdot 24 \left[\frac{h}{d}\right] \cdot 60 \left[\frac{min}{h}\right] - lifetime[y] \cdot annual\ use \left[\frac{programs}{y}\right] \cdot standard\ duration \left[\frac{min}{program}\right]\right) \cdot 10^{-3}}{60 \left[\frac{min}{h}\right] \cdot annual\ use \left[\frac{min}{h}\right] \cdot \left[\frac{programs}{y}\right] \cdot standard\ load \left[\frac{kg}{program}\right]} = passive\ electricity\ use\ [kWh]$$

$$1 \cdot 0.13 \cdot \frac{(12.5 \cdot 365.25 \cdot 24 \cdot 60 - 12.5 \cdot 160 \cdot 108.9) \cdot 10^{-3}}{60 \cdot 160 \cdot 5.92} = 0.014540593327702703$$

5) Ironing

Parameters

$5A = time\ spend\ [min] = 1$
$5B = lifetime\ of\ electric\ kettle\ [y] = 4$
$5C = electricity\ use\ [kWh/min] = 0.027$
$5D = daily\ time\ spend\ on\ ironing\ [min/d] = 5.87$

Activities

Iron impact contribution for 1 min of ironing
This equation calculates the portion of the ironing appliance´s total environmental impact allocated per minute of use.

$$\frac{unit}{lifetime\ of\ electric\ kettle\ [y] \cdot daily\ time\ spend\ on\ ironing\ \left[\frac{min}{d}\right] \cdot 7 \left[\frac{d}{week}\right] \cdot 52 \left[\frac{weeks}{y}\right]} \cdot time\ spend\ [min] = [unit]$$

$$\frac{1}{4 \cdot 5.87 \cdot 7 \cdot 52} \cdot 1 = 0.00011700395005335378$$

Electricity use for 1 min of ironing
This equation calculates the electricity consumed during one minute of ironing.

$$time\ spend\ [min] \cdot electricity\ use\ [kWh/min] = electricity\ use\ [kWh]$$

$$1 \cdot 0.027 = 0.027$$

9) References

This document was written by Emelie Westall Lundqvist in November 2025. Below you find all used input data sources listed, which have also been compiled in the file 'laundry_datasets_references'.

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